laid the intellectual and engineering groundwork for the modern UAVs we know today, blending innovation in aerodynamics, communication, and control mechanisms (Newcome, 2004).

The first practical concepts appeared in the early 20th century, largely driven by military research. During World War I, several nations recognized the potential of unmanned aircraft for reconnaissance and target training. One of the earliest and most influential examples was the “Aerial Target” (1916) developed by A. M. Low in the United Kingdom. This radio-controlled aircraft was among the first attempts to create a pilotless vehicle guided remotely through wireless signals. Although its operational performance was limited, it demonstrated the feasibility of remote-controlled flight.

In the United States, similar developments were taking place. The Kettering Bug (1918), designed by Charles Kettering and funded by the US Army, was an experimental aerial torpedo capable of carrying explosives over pre-set distances using a simple mechanical timer. While it lacked remote control or guidance, it is often considered a precursor to modern cruise missiles and UAVs, representing a key step in the conceptual transition from manned to unmanned aerial systems.

By the 1930s, the evolution of radio technology enabled more sophisticated experiments. The British “Queen Bee” (1935), a radio-controlled target aircraft derived from the de Havilland Tiger Moth, was the first successful operational UAV used for military training. Its success was so influential that the term “*drone*” was coined from its name, inspired by the buzzing sound of its engine, resembling that of a male bee. The Queen Bee demonstrated how remote-controlled aircraft could be used repeatedly and safely for defense training and research, solidifying the concept of UAVs in military practice.